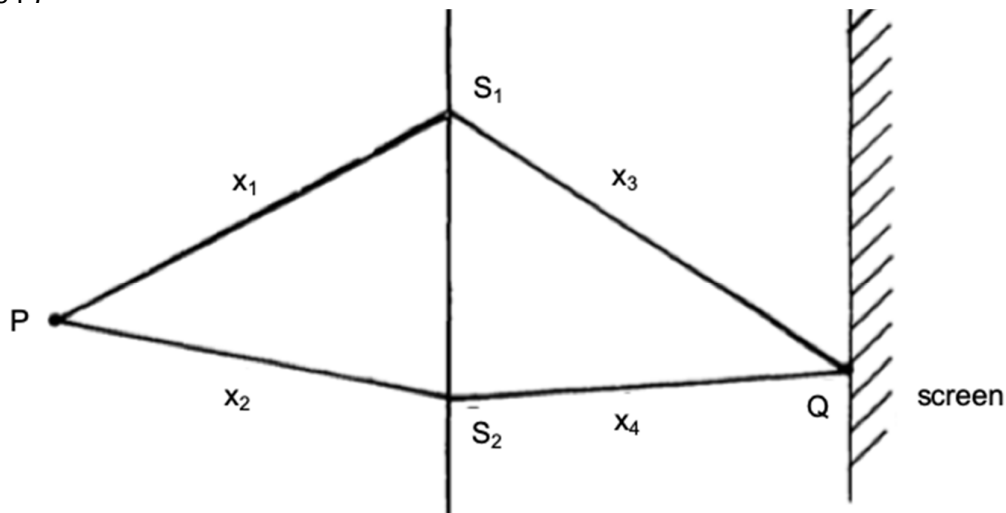


- 19 Two identical narrow slits S_1 and S_2 are illuminated with light of wavelength λ from a point source P.



If, as shown in the diagram above, the light is then allowed to fall on a screen, and if m is a positive integer, the condition for constructive interference at Q is

- A $x_1 - x_2 = (2m+1)\frac{\lambda}{2}$
- B $x_3 - x_4 = (2m+1)\frac{\lambda}{2}$
- C $x_3 - x_4 = m\lambda$
- D $(x_1 + x_3) - (x_2 + x_4) = m\lambda$